

Master's thesis

Master's Programme in Computer Science

Performance, Safety, and Engineering Trade-offs of Rust and C++ in Nokia RAN Control-Plane Systems

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August 24, 2026

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HELSINGIN YLIOPISTO – HELSINGFORS UNIVERSITET – UNIVERSITY OF HELSINKI

Tiedekunta — Fakultet — Faculty	Koulutusohjelma — Utbildningsprogram — Study programme
Faculty of Science	Master's Programme in Computer Science

Tekijä — Författare — Author

Ha Vien

Työn nimi — Arbetets titel — Title

Performance, Safety, and Engineering Trade-offs of Rust and C++ in Nokia RAN Control-Plane Systems

Ohjaajat — Handledare — Supervisors

Keijo Heljanko, Christoph Luneburg, Petteri Kotimaa

Työn laji — Arbetets art — Level	Aika — Datum — Month and year	Sivumäärä — Sidoantal — Number of pages
Master's thesis	August 24, 2026	28 pages

Tiivistelmä — Referat — Abstract

Abstract to be written last.

ACM Computing Classification System (CCS)
Software and its engineering → Software notations and tools → General programming languages
Computer systems organization → Real-time systems → Real-time system specification
Networks → Network types → Mobile networks

Avainsanat — Nyckelord — Keywords

Rust, C++, RAN control plane, tail latency, memory safety, fault injection, incremental adoption

Säilytyspaikka — Förvaringsställe — Where deposited

Helsinki University Library

Muita tietoja — övriga uppgifter — Additional information

Software study track

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1 Introduction

- The 5G RAN control plane (the gNB-CU-CP) now runs as software on ordinary servers. It is large. It keeps state for every connected device and serves many requests at once. The 5G target for control-plane latency, the switch from a battery-saving state to continuous data transfer, is 10 ms [1]. One defect, or one regression in the slowest requests, can degrade the whole site. The work is carried out at Nokia on a UE Context Management service modelled on a real gNB-CU-CP component.

- Today this software is written mostly in C++. C++ gives the control over memory and timing that such a system needs, but the programmer must prevent memory-safety and concurrency mistakes [2]. Rust aims for the same performance while rejecting those mistakes at compile time [3]. The choice matters for a long time. 3GPP has set Release 20 for 6G studies and Release 21 for normative work, with migration from 5G in scope [4]. A language chosen for today's control plane is inherited by the next generation.

- The problem is a lack of evidence. Telecom vendors face growing pressure to move to memory-safe languages [5]. The closest prior work is a cluster of at least five Ericsson master's theses, and none answers the question as posed here. Three keep the runtime in play with the language. They compare Boost.Fibers, goroutines, Tokio, Monoio, Glommio and DPDK Eventdev, not the languages that host them [6, 7, 8]. The other two report execution time and peak memory, not tail behaviour under rising load [9, 10]. Emberg and Wadsten come closest in method. They build a matched C++ and Rust pair joined by a C wrapper, but they measure a stateless transformation [10]. No study holds the specification, concurrency model, and build settings fixed for a stateful, concurrent subsystem, sweeps past saturation, and measures incremental migration cost.

- That evidence is also concentrated in one vendor and untested outside it. On the other side, no study of Rust appears among Nokia's indexed publications at all. The nearest peer-reviewed position argues for making memory-error sanitisation practical, not for replacing it with a compile-time guarantee [11]. That is the

strategy this thesis tests against.

- Four research questions frame the work. RQ1 compares latency predictability (median, p99, p99.9, maximum) under rising load and overload, with the concurrency model held fixed [12]. RQ2 asks which defect classes Rust removes at compile time, at what stage the rest are detected, and which classes it does not prevent. RQ3 asks what overhead Rust's safety model imposes on throughput, CPU, and memory. RQ4 asks what incremental adoption costs in effort, FFI interoperability, and risk inside an existing C++ RAN codebase.

- The work contributes six things. A reproducible, telecom-relevant benchmarking framework. A frozen shared specification with two parity-tested implementations. A comparison of latency and resource behaviour that divides at the saturation knee, where the ordering between the two languages reverses. A defect-class taxonomy that marks where the compile-time guarantee stops, in both directions. A bound on what a comparison of this shape can attribute to a language at all, from decomposing an end-to-end operation into the language's own work and the handoff around it. And evidence-based guidelines for incremental adoption. The dominant cost there is the learning curve, not run-time [13]. The reversal and the attribution bound were both found rather than planned.

2 Background: Systems Languages for Latency-Critical Software

- In the NG-RAN architecture the base station (gNB) is split into a central unit and a distributed unit. The central unit is split again into a control plane and a user plane. UE Context Management sits in the control plane, the gNB-CU-CP, and talks to its neighbours over the standardised F1, E1, and NG interfaces [14].
- Deadlines are met or missed because of predictability, not average speed. The metrics that matter are the p99 and p99.9 latencies and the jitter around them [12]. They must be measured past the saturation point. They must also be recorded without coordinated omission, that is, without letting a load generator that has fallen behind hide its own slowest requests [15, 16].
- C++ offers abstractions that cost nothing at run time, deterministic cleanup through RAII, and no garbage collector. Safety remains the programmer's responsibility [2]. Sanitisers report only the errors that a given test run happens to reach.
- Rust has the same run-time model and also no garbage collector. It checks memory safety and freedom from data races at compile time, using ownership, borrowing, and thread-safety traits such as `Send` [3, 17]. Code that needs to step outside those rules must be marked `unsafe`. That keeps such code small and easy to find.
- The comparison is C++ against Rust. Garbage-collected languages are left out because their pauses are incompatible with the latency targets. The concurrency model is held fixed in both implementations. The C ABI is treated as the realistic adoption path. No earlier study isolates the language this way for a stateful, concurrent subsystem [6, 10].

3 Problem Context: UE Context Management and Predictability

- UE Context Management owns the lifecycle of one context per connected device. Each context moves through four states, Idle, Connected, Reconfiguring, and Released, driven by a small set of lifecycle operations. An operation that is illegal in the current state returns a typed error. It is not carried out.
- The subsystem was chosen because it is representative and still small enough to keep under control. It has mutable state per device, a strict lifecycle, heavy concurrency, and hard latency targets [1, 18]. It is still simple enough to specify precisely and to test for identical behaviour in two languages.
- Seven defect classes fall inside Rust's guarantee. Each has a natural form in this subsystem: use-after-free, double-free, iterator invalidation, data race, unguarded shared mutation, use-after-move, and a thread-unsafe handle shared between workers. The guarantee rests on aliasing control and data-race freedom [17]. The catalogue is this study's own.
- Seven further classes are included on purpose from outside that guarantee: a lock-order deadlock, a lost update, a leak through a reference cycle, an integer overflow in a release build, the blast radius of a failed shard worker, a double release across the C-ABI seam, and a lock guard that is never dropped. They keep the study from confirming a foregone conclusion. They show where the guarantee stops. Two of them return the opposite of the expected result. The C++ tooling reports what Rust's does not.
- Scale is what makes the tail matter. One procedure triggers many operations. One site serves many devices. A p99.9 measured on a single operation becomes a steady rate of degraded procedures in the field [12]. That is why the load sweep is pushed past the point where the service starts to fall behind.

4 Objectives

- A documented literature search of the citation databases and of the Nordic thesis repositories found no study that holds a stateful control-plane subsystem fixed while measuring tail latency of Rust and C++ past saturation under the same concurrency model. The closest work turns out to live in those Nordic repositories [6, 8, 10].
- Five objectives follow from the research questions. O1 measures predictability. O2 establishes which defect classes are prevented. O3 measures the resource overhead. O4 builds and measures a migration prototype. O5 publishes reusable artefacts and guidance.
- Three hypotheses are stated in advance so that the measurements can refute them. H1: below saturation the two implementations perform the same. Prior evidence points to near-parity with a measurable residual, not to equality. A Rust network driver runs 4% slower than its C reference while executing 63% more instructions per packet [19]. H2: Rust prevents at compile time the defect classes for which C++ can only offer best-effort detection. H3: the cost of calling Rust from C++ is negligible.
- The work succeeds if the evidence is good enough to base a decision on. Runs are repeated and reported with confidence intervals [20]. Defect claims are shown on programs that really compile and run. Migration cost is read off a working prototype. Every number is reproducible from the published artefacts.
- The scope is drawn narrowly to protect internal validity. The full gNB, the PHY and L2 layers, the user plane, captured traffic traces, managed languages, and production operations are all left out.

5 Research Methodology

- The approach is empirical and constructive. Instead of comparing two systems that already exist and differ in many ways, the study builds a matched pair of implementations from one frozen specification. The language is then the only systematic difference between them. Both use an identical thread-per-shard concurrency model. That structuring itself changes tail latency [21].
- RQ1 and RQ3 are answered with an open-loop load sweep from light load into overload on pinned cores. Open-loop means the generator keeps to its send schedule instead of waiting for replies. Latency is recorded in identically configured HDR histograms [16]. Resource use is sampled through the operating system. The results so far come from a dedicated dual-socket Xeon host running as a KVM guest. That host resolves the overload divergence and the resource comparison. It does not resolve the far tail below saturation. The frequency governor is not observable from inside the guest and CPU steal is non-zero. Locking frequency requires the bare-metal campaign.
- RQ2 is answered by controlled fault injection [22]. A fixed catalogue of defects is written in both languages. The expected outcome of each one is recorded in advance. A surprise then shows up as a discrepancy, not as something explained away afterwards.
- RQ4 is answered by a working prototype in which C++ drives the Rust core through a foreign-function interface. Cross-language memory-management defects concentrate at that seam [23]. It yields three measurements: how large the seam is, how much `unsafe` code it needs, and what one call across it costs.
- Fairness is the central experimental control. The specification is frozen. Allocator and optimisation settings are matched. Every point where the two languages had to differ is written down in a forced-idiom log. Reporting follows the guidance that a single figure invites conclusions run-to-run variation does not support [20]. Medians come with nonparametric bootstrap confidence intervals [24]. Exact paired permutation tests are corrected for multiple comparisons by the Holm procedure [25]. All data is synthetic, so no data-protection concerns arise.

6 Design and Implementation

- Nothing is implemented until the specification is frozen. It fixes the public API, the UE state machine, the data structures and their algorithmic strategy, the locking and sharding granularity, and the concurrency model. Both implementations conform to it. The comparison then measures the language, not a difference of design or of skill. The specification is itself a deliverable. It is the study's main control against implementation bias.

- One concurrency model is used on both sides, shared-nothing and thread-per-shard. That structuring is known to change tail latency by itself [21]. The identifier space is partitioned into shards. Each shard is owned by a single worker thread that alone reads or mutates it, so no lock guards context state. Callers submit a request to the owning shard's bounded queue and wait on a single-use completion handle. The queue blocks when it is full. That is deliberate. A drop policy would hide overload, and this study needs overload to appear as latency. The only shared mutable state either version permits is one atomic counter. No control flow depends on its value.

- Fairness is enforced below the design as well as in it. Both builds link the same `jemalloc` and keep a system-allocator build as a control. Both compile at matched optimisation. Both record latency through one HDR histogram algorithm configured identically [16]. Hashing is pinned on both sides, SplitMix64 for shard assignment and FNV-1a inside the map. C++'s default hash of an integer is typically the identity function. Rust's is cryptographically strong. Left as they came, the two standard libraries would have been the measurement.

- Where the languages cannot satisfy a requirement the same way, the divergence is written into a forced-idiom log and then measured. Rust takes its bounded blocking queue from `sync_channel`. C++ through C++23 has no such primitive. It assembles one from a mutex and two condition variables. C++ has no payload-carrying enumeration in this idiom either. A request becomes a tagged struct holding every argument field any operation might need, 160 bytes against Rust's 96. The caller answers through a `variant` it has to unwrap. The net result is

a Rust core 18% longer in code lines whose total cyclomatic complexity is 15% lower [26]. The branchiest function in either core is the C++ worker loop that dispatches that `variant`. The Rust library contains no `unsafe` at all.

- Parity is tested and not assumed. Language-neutral vectors run in both implementations and their transcripts are diffed operation by operation. A cross-diff cannot tell “both correct” from “both wrong in the same way”. A third implementation in Python acts as an independent oracle. It has to reproduce all 223 hand-derived expectations before it may generate any of its own. A campaign of roughly 793 000 generated operations found no divergence. The transition matrix is enumerated exhaustively instead of sampled. The C++ side runs under AddressSanitizer, ThreadSanitizer and UndefinedBehaviorSanitizer. The Rust side runs under Miri. Line coverage is reported. A clean sanitiser result is therefore bounded by the code that actually ran.

7 Experimental Setup and Evaluation

- The workload is synthetic and spec-justified. A deterministic driver walks each UE through the connection lifecycle the 3GPP specifications define. Its draw order is pinned, so the per-operation-kind counts come out identical in both languages for a given seed. That doubles as a parity signal. Each sweep point is repeated 20 times. The four language and allocator variants are interleaved inside every repeat, so thermal or background drift falls on all of them alike. Every point is a median with a bootstrap 95% confidence interval [20, 24]. The representative operating point is not chosen by eye. It is the busiest swept load at which all four variants still hold p99 below 5 ms, 56 kops/s against a measured saturation knee of 335 kops/s. All of it runs on a dedicated dual-socket Xeon pinned to cores 2–15. That host bounds what the campaign may claim, because it is a KVM guest. The frequency governor and the turbo state are unobservable from inside it, so frequency stability is assumed instead of enforced. CPU steal is non-zero at roughly the order of the latencies under study. Run-level statistics survive that. The overload and the resource comparisons are resolved here. The far tail below saturation is not. It is reported as unresolved, not as a null. The bare-metal protocol that would settle it is specified and published.

- Below saturation the four variants sit within $2\,\mu$s of one another at the median, 2.1%. The paired test resolves a difference of $1.6\,\mu$s in Rust's favour under both allocators. Higher up the distribution the instrument runs out. The widest interval on a single variant's own p99 spans $1480\,\mu$s against $293\,\mu$s of spread between all four. Where the uncertainty about one variant exceeds the gap between them, an ordering would be reporting the scheduler. A second limit is structural. No host would fix it. An end-to-end operation is 0.250% the work the language performs. The rest is a cross-thread handoff charged to the operating system [27]. This sweep cannot bound a language effect in either direction. Measured on its own the bare operation is 1.6× faster in Rust. The likely cause is the standard hash map's layout, not anything about safety.

- Past saturation the languages are not equivalent. The margin is not marginal. At 784 kops/s offered, C++ sustains 116 kops/s where Rust sustains 63, a factor of 1.8. That holds in 20/20 repeats under both allocators, with the Holm-corrected p at the floor of the whole test family. The slower variant is not simply doing less work. It draws 1.9 times the CPU per completed operation. That is the signature of effort spent without completing anything. Isolating the bounded queue and the completion handle on their own does not reproduce that ordering at significance. The mechanism sits in the wider handoff path. This is the largest and most reproducible language effect in the study. A comparison that stopped at nominal load would have concluded the opposite.

- Safety is measured by injecting 14 defects. Seven come from inside the region Rust claims to govern. Seven were chosen because safe Rust is expected *not* to prevent them [22]. Every one of the first seven is a compile error in Rust. It cannot be written at all. In C++ each is valid code. The defect appears only if a test drives it, and no single sanitiser covers them. The use-after-move is the case where no sanitiser reports anything: a moved-from object is valid but unspecified, so there is nothing to detect and the identity is simply lost. All seven boundary cases compile in safe Rust and misbehave. On two of them the C++ tooling is the one that reports. LeakSanitizer flags the reference cycle that Rust leaks in silence. UndefinedBehaviorSanitizer flags the overflow that Rust defines to wrap. ThreadSanitizer catches the lock-order inversion on some runs and not others, at a rate that is itself unstable. It builds its lock-order graph from acquisitions that succeed. A run whose first inversion blocks never closes the cycle. Two Rust model checkers do report both original boundary defects, but only after each was re-imported against the checker's own primitives and shrunk to model size [28]. Miri reports the seam double release, the one class that safe Rust rejects and unsafe Rust does not, but it interprets Rust and cannot execute the C++ host the defect ships behind. That half of the boundary is caught at test time, and for the concurrency cases only after a rewrite.

- Resource use separates the variants where latency could not. It holds two effects of unequal size. The Rust build costs 1.9–3.4 MiB more peak resident set than the C++ build under either allocator. Linking `jemalloc` at all costs 32.9–34.4 MiB in either language. Both are real. The allocator's effect is an order of magnitude the larger. The language difference is a fixed cost, not a per-context one. No

hardware counters were collected, so it is established without being explained. CPU inverts with the regime. It is marginally lower for Rust while the service copes and much higher per completed operation once it does not. The migration seam is small on every axis measured: 11 C entry points in 231 lines of Rust. All raw-pointer work is confined to 9 functions. The core is left free of `unsafe`. No cost is detected against the same operation called natively. That bounds any per-operation tax at 10.91%. Three adoption costs beyond the seam were measured for the first time in this pass. Two contradict the received objection. 54% of the Rust clean build is third-party code. The project's own source compiles in 0.8 times the CPU `g++` needs. The 6.8× binary-size gap is a linkage policy that reverses when the policy is held fixed. The compared component resolves 0 crates and contains 0 `unsafe` sites.

8 Discussion

- RQ1's answer divides at saturation. Reporting either half on its own would misrepresent it. Below the knee the two languages are equivalent in central behaviour by a margin no deployment would notice. That is a result about services of this shape. The far tail in the same regime stays unresolved. That is an admission about the instrument. Past the knee they diverge by 1.8×. The closest prior work measures this quantity for 5G control-plane procedures. It finds the same shape and traces it to stackless `async/await` [8]. That attribution is accepted here. The shape appeared with no asynchronous runtime on either side and the concurrency model held fixed. A runtime cannot be the only place such an effect originates. The prediction registered in advance also named where to look first, bounds and overflow checking. That part of it was wrong. Recording the miss is worth more to the next study than quietly substituting the explanation that survived.

- On safety the difference is one of kind. The boundary is reported as a first-class result, not as a caveat. A compiler that refuses to build a program rules the defect out of every execution. A sanitiser rules it out only of the executions an instrumented test happens to take. What is absolute there is the execution coverage, not the breadth of the catalogue. Rust reasons about aliasing and not about acquisition order. A `Mutex` guard whose lifetime the borrow checker tracks perfectly says nothing about whether two threads take their locks in a compatible order [29]. The lost update escapes ThreadSanitizer for the same reason it escapes the borrow checker. Where every access is correctly synchronised, there is no unsynchronised access to find. Adopting Rust retires one category of concurrency defect and leaves untouched the category that fails silently. A migration plan reading "memory-safe" as "concurrency-safe" would mis-budget its review effort. One case reaches past the injected programs and narrows the credit due on both sides. A specification gap let an operation be submitted outside the service's running window. It survived every test, every parity vector and all three sanitisers because nothing ever called that path. That is the weakness of detection conditional on coverage. What turned undefined behaviour into a diagnosable

failure on the Rust side was an ordinary bounds check, not a proof. The property worth budgeting for is “checked by default”, not “ruled out at compile time”. The root cause belongs to the method. A prohibition stated without a mechanism behind it leaves each implementation to fail in its own idiom.

- RQ3 splits by resource and the splits do not point the same way. RQ4 comes out cheap in the machine and expensive in the people. The memory penalty reproduces under both allocators. It is credible beyond its p-value. Peak resident set is a run-level maximum insensitive to the jitter that destroys a p99. It is a few MiB of fixed cost against an allocator effect an order of magnitude larger. One close study reports the opposite sign, with Rust lowest and the gap widening as concurrency rises [6]. The two are compatible. Their C++ runs on Boost.Fibers, which allocates a stack per fiber. Their Rust runs on Tokio, where a task is a small heap-allocated future. What they measured is the per-context cost of two concurrency runtimes. Two further studies agree with the direction found here [9, 30]. On migration the dominant cost is human [13, 31]. Confining `unsafe` to the seam bounds what has to be reviewed, not what can be corrupted. Host and component share one address space [32, 33]. The build and footprint axes point the same way as the seam. An organisation worried about either should audit its dependency budget before blaming the compiler. The one axis where C++ leads outright is the edit-compile loop for changes that stay inside a translation unit.

- The three hypotheses were committed before any measurement. Stating their disposition is what separates a test from a description. H1 is supported below saturation and refuted above it. It is not in the direction its wording anticipated. The difference appeared in overload, not in the far tail H1 actually named. It was also not well posed. “Performance-equivalent” reads as one quantity and names at least three. H2 holds without exception for the seven injected classes of its kind. It fails in the looser reading its wording invites. It is accepted for memory safety and data-race freedom, rejected for concurrency correctness at large, and it is a claim about *safe* Rust only: the seam case shows the same class returning inside `unsafe`. H3 is supported by the bound on a cost the measurement could have missed, not by the failure to detect one. Three threats stay load-bearing behind all of it. One author wrote both implementations. The parity suite bounds behaviour and not performance [10]. The end-to-end instrument measures the design faithfully and the language barely at all. The overload

divergence is replicated here with its mechanism narrowed and not closed, and no peer-reviewed study reports the same reversal.

- The guidelines are the practical form of all this. Each carries the limit of the finding behind it. Choose a first component for the narrowness and the scalarity of its interface, not for its size. The redesign of a rich interface is the schedule risk. The boundary itself is not. Keep the C ABI in its own crate and the core free of `unsafe`. That makes the reviewable surface a countable thing [34]. Leave the concurrency review budget where it is. Prefer a shared-nothing structure in either language. The strongest concurrency result here belongs to the design and to neither compiler. Plan the learning curve as a training cost and instrument it. The second component can then be scheduled on measured effort. Measure the component as well as the round-trip. Sweep past the point where the service stops coping. That is the guideline this study would most like to have followed from the start.

9 Conclusion and Future Work

- The thesis compared modern C++ and Rust on a UE Context Management service modelled on a real gNB-CU-CP component. It looked at four dimensions: tail-latency predictability, compile-time defect prevention, safety-model overhead, and incremental-adoption cost. Both implementations were built against one frozen specification and driven along identical code paths. That is what lets a language effect be told apart from a difference of design or of skill.
- On safety the difference is one of kind and it has a definite edge. The edge stops where the catalogue says it does. Safe Rust rejects all seven injected memory-safety and data-race classes at compile time. Hazards that C++ can only hope to catch with instrumented tests become programs that cannot be written. All seven boundary classes compile and misbehave in safe Rust as they do in C++. On two of them the C++ tooling reports what Rust's does not. Stated precisely, Rust eliminates data races and aliasing errors. It does not eliminate race conditions, lock-ordering errors, leaks, wrapping arithmetic or the blast radius of a failed worker, and it eliminates nothing inside an `unsafe` block.
- On performance the evidence divides twice before it divides by resource. Getting both divisions right was this study's most consequential correction to itself. It divides first by what is being measured. An end-to-end operation here is 0.250% language. The rest is an operating-system handoff. An end-to-end null bounds nothing about either compiler. The same operation measured apart is 1.6× faster in Rust. It divides second by whether the service is coping. Below saturation the two are equivalent at the median for any purpose a deployment would care about. The paired test does resolve 1.6 μs in Rust's favour, so that equivalence is practical and not statistical. Their far tail is unresolvable on a virtualised host. Past saturation C++ sustains 1.8× the throughput of Rust while drawing 1.9× the CPU per completed operation. On memory the Rust build carries a few MiB more fixed footprint under both allocators. That is an order of magnitude less than the allocator choice costs.
- What follows is a narrow claim, narrower than this thesis expected to make. An unmodified Rust core can be driven from C++ across a thin seam. The cost

on a real operation is bounded at 10.91%. All `unsafe` is confined to the seam. The core is verified free of it. For stateful, concurrent, partitionable control-plane services of this kind, and at nominal load, the choice between the two languages is not a performance decision. It is a decision about which classes of defect an organisation wants to make unwriteable, and about whether it will pay the engineering and learning cost of getting them. Under overload it becomes a performance decision again. It currently favours C++. What the thesis adds is an instrument, not a verdict: a frozen specification, a parity-tested pair of implementations, a measurement framework that reaches past the saturation knee, and a defect catalogue that maps where the guarantee stops.

- Future work answers the limitations one at a time. The immediate priority is the bare-metal campaign. Its target is now narrow: the far tail below saturation, with hardware counters and allocation profiles collected on the same visit. That is what would turn the memory result from an observation into an explanation. Beyond it, replaying anonymised production traces would validate the operation mix. A subsystem with genuinely shared mutable state would exercise the two languages differently than a shared-nothing design does. Evaluating Rust async runtimes against the thread-per-shard model would separate the runtime from the language in the tail result [8]. Measuring what in-process isolation of the two heaps costs would price what protecting an `unsafe`-free core is worth [32]. A longer case study tracking defect density and review effort during an incremental adoption is what would settle the human cost this study can only name.

Statement on the Use of AI Tools

Tool. Anthropic's Claude Opus 5, used through the Cursor development environment, is the only generative AI system used in preparing this thesis, and it was used throughout, from June to November 2026. It is named neither as an author, which University of Helsinki policy forbids, nor as a source, which is the author's own choice.

What the assistance covered. It was extensive. The model drafted and revised the running prose and proposed the order in which the arguments are made. It wrote and refactored both implementations of the UE Context Management service, the load generator, the fault-injection catalogue, the interoperability prototype, the analysis code behind the bootstrap intervals and the paired permutation tests, and the figure-generating, build and LaTeX sources. It also suggested candidate literature. The abstract is the one part it must not touch: University of Helsinki policy never permits a language model in a maturity test, and in this programme the maturity test is the abstract. In this draft it has not yet been rewritten unaided, and this statement will say so until it has been.

What the model did not settle. The research problem, the four research questions, the three hypotheses, and the control on which the study's value rests are the author's. That control is one frozen language-neutral specification and one concurrency model, so language remains the only systematic difference between the two implementations. The author also chose to carry defect classes from outside Rust's compile-time guarantee, which is what keeps RQ2 falsifiable. Every measurement was run by the author on the hosts documented here, and the quantities the thesis quotes reach the page as macros regenerated from the raw logs rather than as figures typed into a paragraph. The readings are the author's too, including the two restraints the evidence imposes: an unresolved far tail is not equivalence, and a sweep dominated by cross-thread handoff cannot bound an effect of the language.

How the output was checked. Model output was treated as a draft to be falsified. Every bibliography entry was resolved against Crossref, DataCite, or the issuing body and traced to the page or clause carrying the claim made of it. That audit found four passages that overstated or misplaced their source, one a quotation absent from the work it was attributed to. All four were corrected, and the audit is kept beside

the artefacts. Code was checked by running it, against a conformance suite whose transcripts are diffed line by line and a fault catalogue that fixes each expected outcome before the defect is injected.

What the author took from it. The model is dependable at mechanical work, at transposing an idiom between two languages or keeping a build honest. It is undependable in one predictable way: it will supply a plausible citation for a claim and a confident sentence where the data supports a hedge. Its fluency is not evidence. The effort saved in drafting is spent again in checking, and the defensible parts of this thesis are the parts checked against a source, a specification, or a measurement.

Responsibility. The author is responsible for the whole of this thesis. Where it is wrong, the error is the author's.

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